



APRIL 2024

REGIONAL MARITIME UNIVERSITY

DEPARTMENT: MARINE ENGINEERING

PROGRAMME: DME 25 BME 27 BCE27/BMT27 DEE25

COURSE: WORKSHOP TECHNOLOGY 1

END OF SEMESTER EXAMINATIONS

INSTRUCTION

Answer all questions section A (objectives) and section B (theoretical questions)

Time: 2hrs

SECTION A

2 MARKS EACH

1. A gimlet is a - tool
 - a. Boring
 - b. Marking
 - c. Cutting
 - d. Parting
2. Scribbling block is used to
 - a. Hold round bars during marking
 - b. Check trueness of flat surfaces
 - c. Locate the centers of round bars
 - d. Check the surface roughness
3. The cutting edge of chisel should be
 - a. Hardening
 - b. Tempered
 - c. Case hardened
 - d. Hardening and tempered

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4. A ring gauge is used
- Check the diameter of shafts and studs
 - Test the accuracy of holes
 - Check the clearance between two mating surfaces
 - All the above
5. A hacksaw blade is specified by its
- Length
 - Material
 - Width
 - Number of teeth
6. For cutting brass the hacksaw blade should have -----per 25
- 14
 - 24
 - 32
 - 40
7. A file removes metal during
- Forward stroke
 - Return stroke
 - Both forward and return stroke
 - None of the above
8. Notching is the operation of
- Cutting a sheet of metal in a straight line along the length
 - Removal of metal to the desired shape from the edge of the plate.
 - Bending sheet of metal along curved surface
 - Cutting a piece of metal through its length
9. The teeth of a hacksaw blade are bent
- Towards right
 - Towards left
 - May be bent towards any direction
 - Alternatively towards right and left and every third or fourth left straight
10. V Block is used to
- Check the trueness of flat surfaces
 - Locate centers of round rods
 - Check the surface of roughness

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- d. None of the above
11. The cross-section of a chisel is usually
- Rectangular
 - Square
 - Hexagonal
 - Octagonal
12. The instrument used to measure external and internal diameter of shafts , thickness of parts and the depth of holes, is
- Outside micrometer
 - Inside micrometer
 - Depth gauge micrometer
 - Vernier caliper
13. Which of the following materials is non ferrous
- Mild Steel
 - Aluminum
 - Brass
 - Naval brass
14. Which of the following tools is used for striking
- Chisel
 - Hammer
 - File
 - Punch
15. Which of the following tool is used to originate a hole?
- Nail
 - Drill
 - Bore
 - Hammer
16. ---- is a personal safety equipment.
- Ruler
 - Hacksaw
 - Helmet
 - Relay
17. Oil spillage at a workshop may affect
- Personal safety
 - Environmental safety

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- c. Equipment safety
- d. All the above

18. Which of the following do not have a ferrous metal

- a. Cast irons, brass, bronze
- b. Naval brass, aluminum, steel
- c. Cast iron, wrought iron, Steel
- d. Copper, brass, bronze

19. In order to cut a mild steel the cutting angle of the chisel should be

- a. 60°
- b. 45°
- c. 50°
- d. 45°

20. Which of these is not a composite material

- a. Wood
- b. Bone
- c. Thermosetting
- d. Fiberglass

SECTION B

(1) As a student engineer you visited your village explain engineering materials to them
30 MARKS

(2) i. Briefly explain the process of casting.

8 MARKS

ii. Discuss conditions for selecting engineering tools.

7 MARKS

(3) i) with the aid of diagram describe micrometer screw gauge and explain the method of its usage.

6 MARKS

ii) list and discuss the safety methods you know and give 3 examples each.

9MARKS